

Online Resource 1

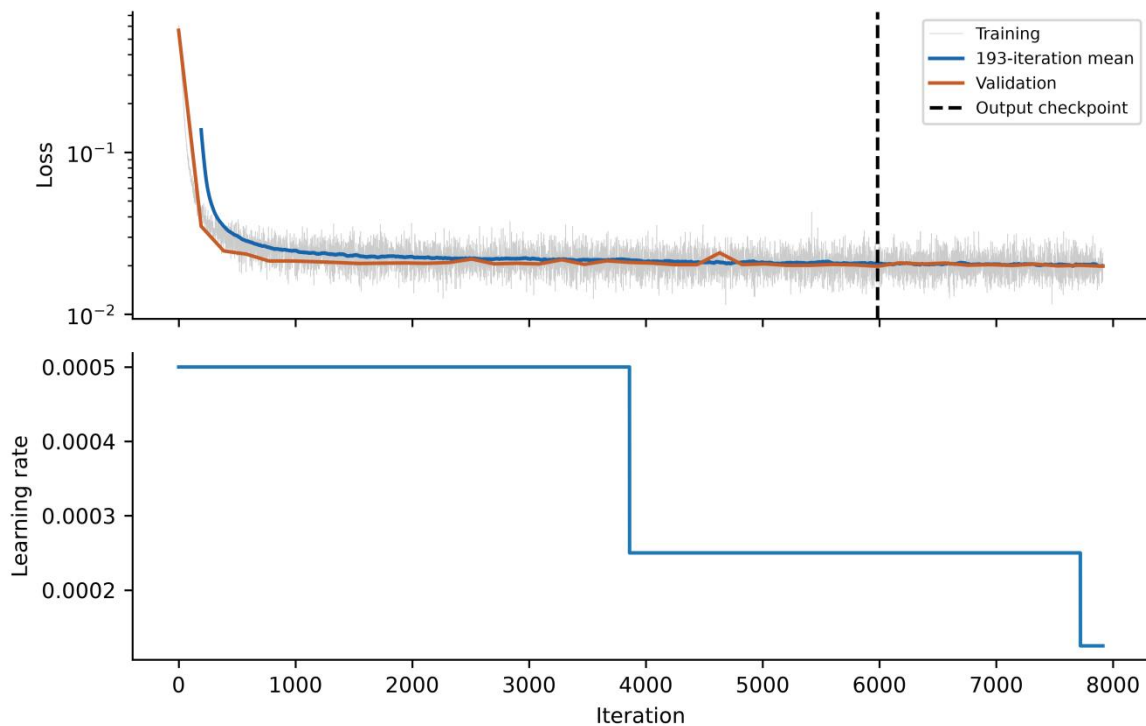
Waveform Identity and Reproducible Source-Grouped Evaluation of Seismic Phase Pickers

Earth Science Informatics

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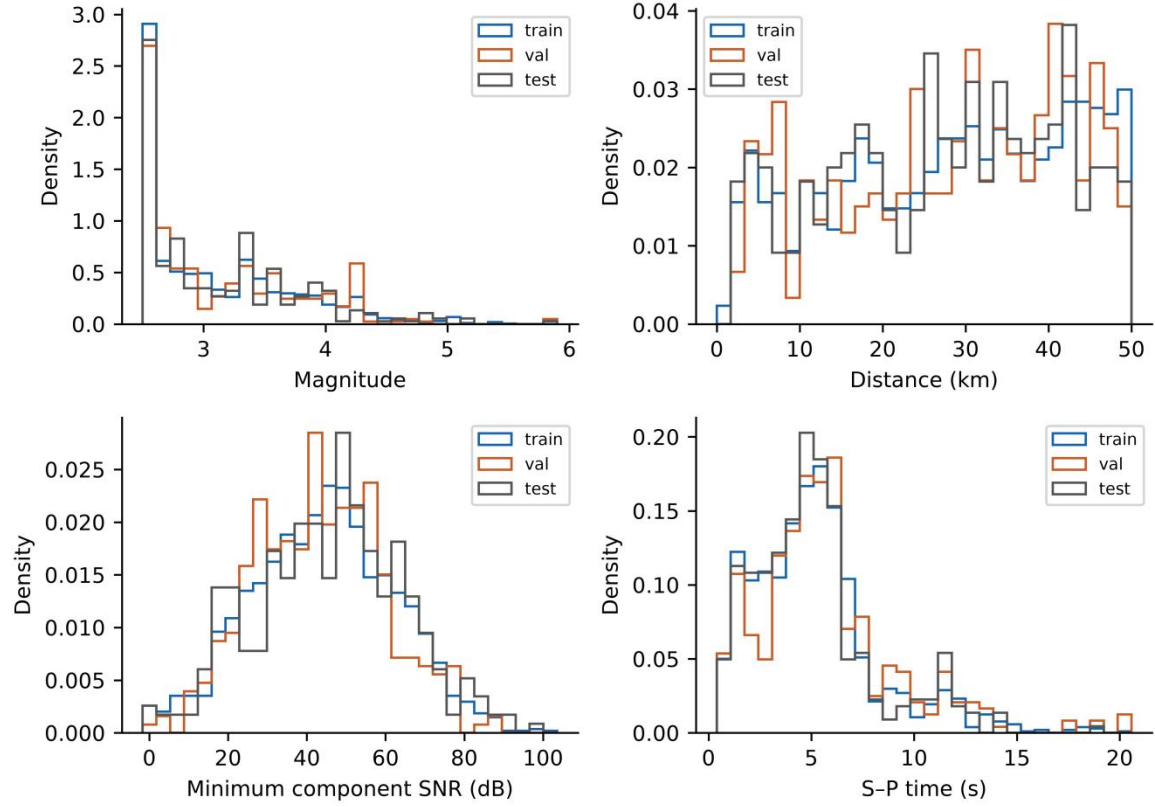
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Scope of results. Empirical tables and figures use the corrected source-verified v2 run. Base-42 diagnostics vary meta seeds 42–44; expanded summaries cross base seeds 42–44 with those meta seeds. Fresh inference replay verifies the current artifacts. Historical data roles, manual-status sensitivity and station/threshold diagnostics qualify predictive interpretation. This Online Resource accompanies the ESI manuscript; its tables and figures retain the verified v2 numerical content.

Supplementary Fig. S1. Current Full3C base-42/meta-42 training and validation loss. The moving mean spans 193 iterations (one epoch). Output network iteration was 5983; training continued to iteration 7913. The learning-rate schedule is shown below.



Supplementary Fig. S2. Recovered magnitude, distance, minimum component SNR and S–P time distributions by subset. Source disjointness is verified from identifiers, not from these distributions.

Supplementary Table S1. Statistics over all 330 current Full3C test tensors (6000 samples each; 1,980,000 time samples per channel). Sample SD uses $n-1$.

Ch.	Feature	Min	Max	Mean	SD	NaN / Inf
1	P_STA	0	1	0.0062666	0.066271	0 / 0
2	S_STA	0	1	0.0062666	0.066271	0 / 0
3	Noise_STA	0	1	0.99219	0.075265	0 / 0
4	P_AIC	0	1	0.0041777	0.054191	0 / 0
5	S_AIC	0	1	0.0041777	0.054191	0 / 0
6	Noise_AIC	0	1	0.99389	0.066808	0 / 0
7	P_CNN	1.0103e-08	0.98778	0.0024867	0.028303	0 / 0
8	S_CNN	2.6443e-12	0.9966	0.0029396	0.015484	0 / 0
9	Noise_CNN	0.0034005	0.99999	0.99457	0.032189	0 / 0
10	P_TCN	2.9367e-09	0.98743	0.0024009	0.032685	0 / 0
11	S_TCN	1.8046e-11	0.97655	0.0029954	0.024957	0 / 0
12	Noise_TCN	0.012574	1	0.9946	0.041079	0 / 0
13	E	-30.302	36.117	4.203e-12	0.99992	0 / 0
14	N	-36.129	42.767	1.9321e-11	0.99992	0 / 0
15	Z	-34.743	25.54	7.8224e-12	0.99992	0 / 0

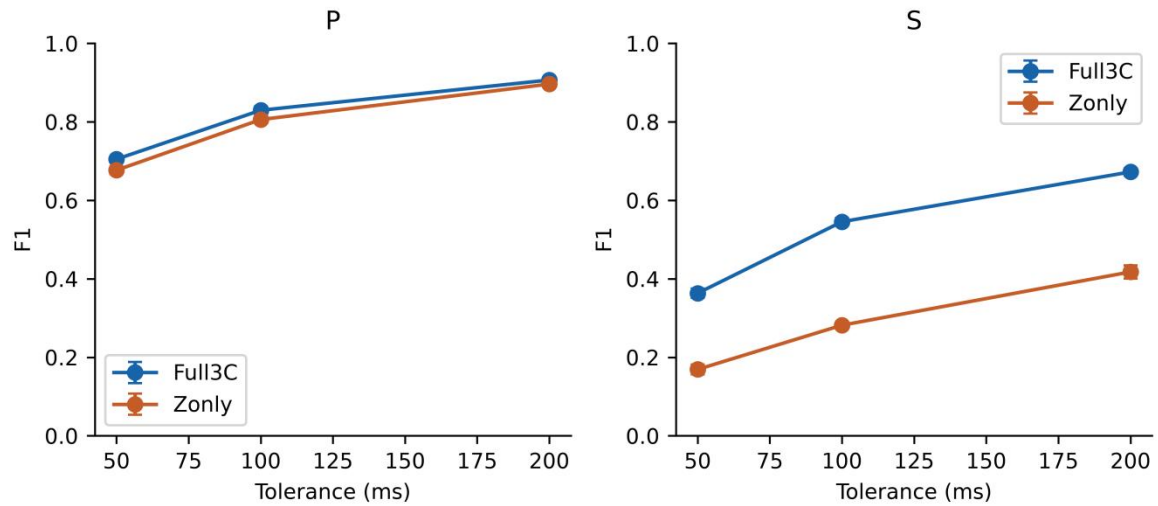
Supplementary Table S2. Three-seed mean F1 and percentile 95% source-cluster intervals, with paired Full3C minus Z-only differences.

Mode / comparison	Phase	Tolerance ms	Mean F1 / Δ	95% CI
Full3C	P	50	0.7049	[0.6427, 0.7627]
Z-only	P	50	0.6768	[0.6121, 0.7380]
Full3C minus Z-only	P	50	0.0280	[0.0030, 0.0555]
Full3C	P	100	0.8296	[0.7807, 0.8753]
Z-only	P	100	0.8059	[0.7554, 0.8539]
Full3C minus Z-only	P	100	0.0237	[-0.0012, 0.0510]
Full3C	P	200	0.9067	[0.8701, 0.9394]
Z-only	P	200	0.8963	[0.8592, 0.9296]
Full3C minus Z-only	P	200	0.0104	[-0.0032, 0.0254]
Full3C	S	50	0.3634	[0.3063, 0.4186]
Z-only	S	50	0.1691	[0.1283, 0.2108]
Full3C minus Z-only	S	50	0.1942	[0.1380, 0.2507]
Full3C	S	100	0.5455	[0.4803, 0.6074]
Z-only	S	100	0.2818	[0.2274, 0.3347]
Full3C minus Z-only	S	100	0.2637	[0.2046, 0.3231]
Full3C	S	200	0.6725	[0.6122, 0.7298]
Z-only	S	200	0.4176	[0.3538, 0.4817]
Full3C minus Z-only	S	200	0.2549	[0.1959, 0.3155]

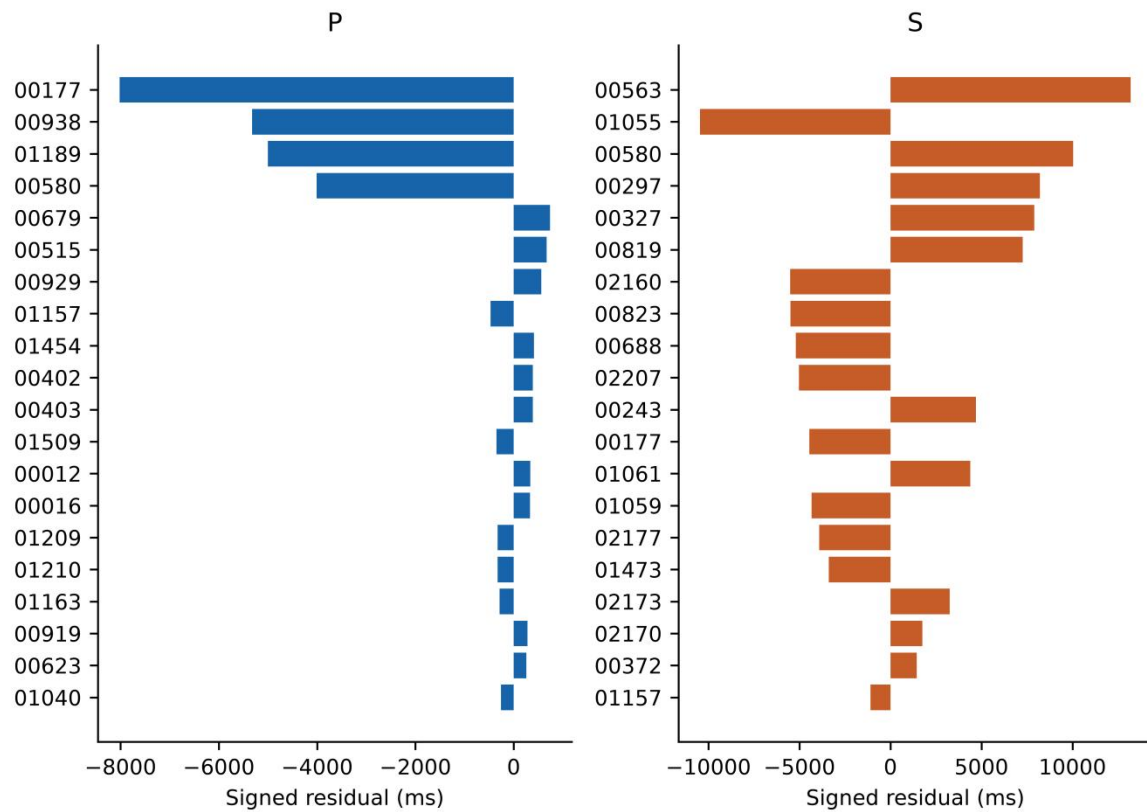
The 221 test sources were resampled with replacement, retaining all records from each selected source. We used 10,000 NumPy default_rng draws with seed 42. Each draw recomputed per-fit F1 before averaging the three meta fits. Modes share source draws. Intervals condition on the fixed models and split; they exclude training and split uncertainty.

Supplementary Table S3. Seed-42 absolute-error statistics for accepted (detected or uncertain) picks. Percentiles use linear interpolation; outlier percentages use accepted denominators. These single-seed diagnostics accompany Supplementary Table S29 three-seed summaries.

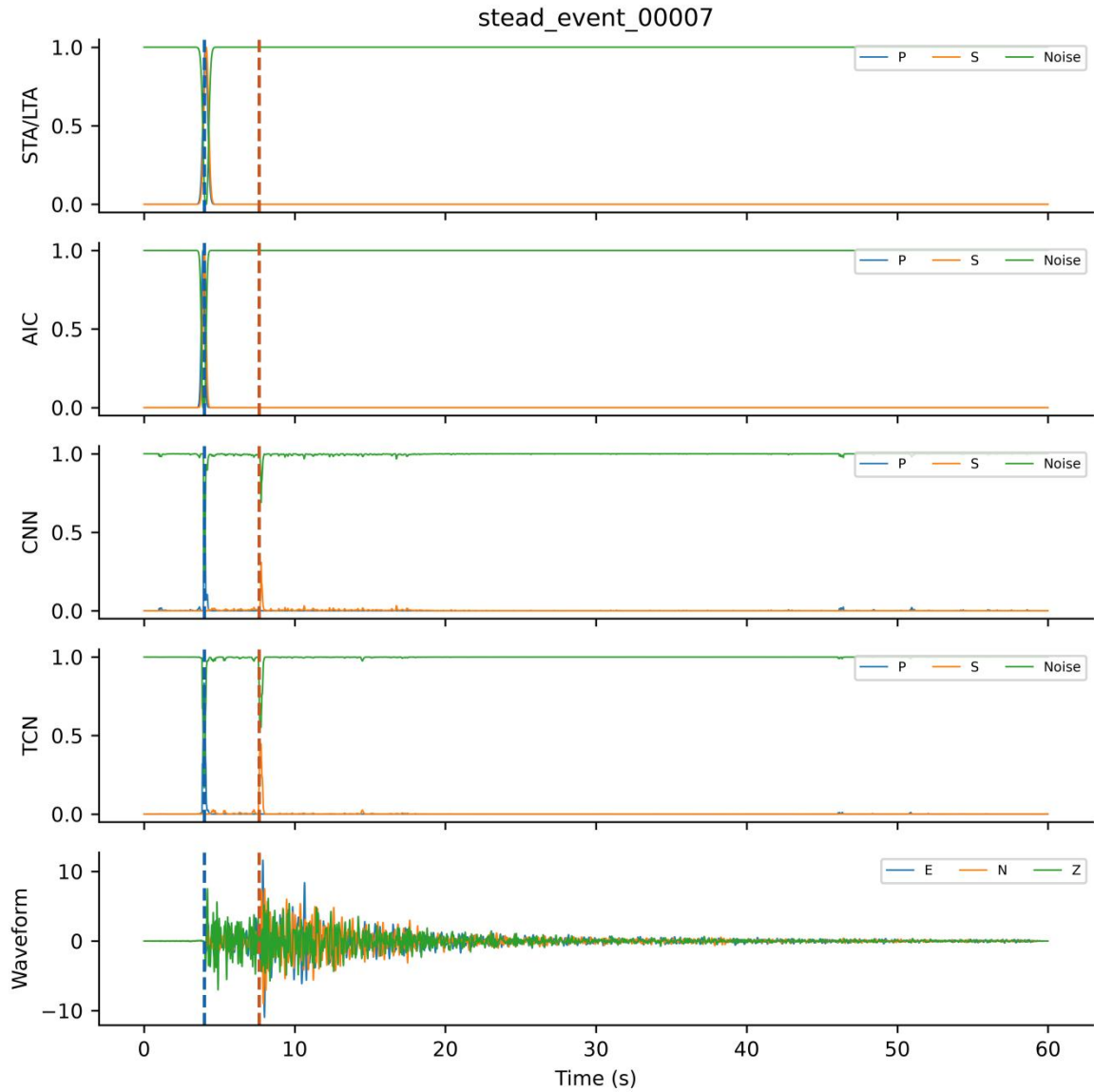
Metric	Full3C P	Full3C S	Z-only P	Z-only S
Accepted picks	327.00	322.00	328.00	315.00
MAE (ms)	126.97	492.20	169.54	1170.00
Median AE (ms)	30.00	80.00	30.00	250.00
RMSE (ms)	649.26	1596.78	947.78	2532.54
P90 (ms)	160.00	629.00	203.00	3980.00
Error >500 ms (%)	2.14	13.35	2.44	34.60
Error >1 s (%)	1.22	6.21	1.52	22.86
Error >2 s (%)	1.22	5.28	1.52	16.19



Supplementary Fig. S3. Mean F1 versus tolerance. Error bars show sample SD over three meta-learner seeds, not source-bootstrap intervals.



Supplementary Fig. S4. The 20 largest absolute P and S residuals among Full3C seed-42 accepted picks, with signed errors and event-ID suffixes. This is a ranked diagnostic, not an exhaustive waveform panel; selected waveforms appear in Main Fig. 7.



Supplementary Fig. S5. Current final Full3C test tensor for `stead_event_00007`. Twelve P/S/noise channels precede conditioned E/N/Z. Dashed lines show true arrivals; Table S1 covers all 330 test tensors.

Supplementary Table S4. Final settings for the completed source-verified 40-fit main workflow. Supplementary Table S28 provides architecture and optimizer details. Logistic uses independent binary phase heads and does not isolate nonlinearity.

Category	Parameter	Value
Dataset	Population	2234 records / 1477 sources
Dataset	Sampling; duration	100 Hz; 6000 samples; 60 s
Curation	Distance; magnitude; SNR	Recovered population; see Main Table 1
Curation	Reference arrivals	1817 records both manual; 417 include automatic/autopicker
Partition	Train sources / records	1034 / 1544

Category	Parameter	Value
Partition	Validation sources / records	222 / 360
Partition	Test sources / records	221 / 330
Partition	Source intersections	Zero for every pair
OOF	Grouping; folds; inner validation	source_id; 5; 15% of non-held sources
Meta input	Augmented training records	3088 per mode
Meta input	Full / prob-only / waveform-only	6000×15 / 6000×12 / 6000×3
Training	Completed fits	40 (25 Full3C; 15 Z-only)
Seeds	Base / neural meta	42 / 42,43,44
Augmentation	Noise; amplitude range	20 dB; 0.5–2.0
Augmentation	Horizontal dropout	Probability 0.20
Augmentation	Time shift / polarity flip	Disabled / disabled
Logistic	Heads; target	P,S binary; phase mask >0.3
Logistic	Normalization	OOF-training mean and sample SD
Logistic	Subsampling / ridge penalty	Every 10th sample / 0.01
Logistic	Optimization	Newton; ≤100 iterations; gradient∞<1×10 ⁻⁶
Evaluation	Accepted / strict masks	Detected+uncertain / detected; finite
Evaluation	Tolerances	±50, ±100, ±200 ms
Evaluation	MAE denominator	All accepted, including outliers
Bootstrap	Mean intervals	10,000 source draws; NumPy seed 42
Software	Training	MATLAB R2024a
Software	Tables / figures	Python; NumPy, pandas, Matplotlib
Provenance	Frozen results	source-verified v2 / submission40
Provenance	Manifest at audit	25,248 manifest entries matched across all runs

Supplementary Table S5. Full3C F1 on 330 records. Bases use final seed-42 fits; I-CNN entries are three-seed means (SD in Table S6). All methods use the primary decoder and acceptance policy. Weighted coefficients derive from phase-specific validation F1@100 ms, fixed before test evaluation.

Method	P50	P100	P200	S50	S100	S200
STA/LTA	0.2152	0.4576	0.7879	0.0062	0.0154	0.0463
AIC	0.2848	0.3697	0.4364	0.0000	0.0000	0.0000
CNN	0.6881	0.7890	0.8716	0.2595	0.3599	0.4533
TCN	0.7003	0.8226	0.9021	0.3506	0.4935	0.6558
Mean ensemble	0.6818	0.7879	0.8667	0.1131	0.1407	0.1590
Validation-weighted	0.6848	0.7939	0.8667	0.3551	0.4958	0.6298
Stacker mean	0.7049	0.8296	0.9067	0.3634	0.5455	0.6725

At base seed 42, TCN P/S F1@100 ms was 0.8226/0.4935 versus stacker means 0.8296/0.5455. The S difference had an exploratory paired interval of 0.0165–0.0902; the P interval included zero. This result concerns the current fixed split and these comparators. PhaseNetMatched appears separately in Table S16.

Supplementary Table S6. Full3C ablation. Neural variants report mean $\pm$ sample SD over seeds 42–44 on shared OOF features; logistic was fitted once. The last column gives control minus variant Δ F1@100 ms and paired source-cluster 95% interval (10,000 draws). These multiple intervals are exploratory and not multiplicity-adjusted.

Variant	Phase	F1@50 ms	F1@100 ms	F1@200 ms	Control – variant Δ and 95% CI
Full15 control	P	0.7049 $\pm$ 0.0033	0.8296 $\pm$ 0.0093	0.9067 $\pm$ 0.0060	—
Full15 control	S	0.3634 $\pm$ 0.0121	0.5455 $\pm$ 0.0099	0.6725 $\pm$ 0.0086	—
Probability12	P	0.6947 $\pm$ 0.0094	0.8327 $\pm$ 0.0051	0.9087 $\pm$ 0.0082	-0.0030 [-0.0162, 0.0105]
Probability12	S	0.3534 $\pm$ 0.0026	0.5177 $\pm$ 0.0143	0.6646 $\pm$ 0.0099	+0.0278 [0.0056, 0.0523]
Waveform3	P	0.6762 $\pm$ 0.0097	0.7957 $\pm$ 0.0080	0.8815 $\pm$ 0.0033	+0.0339 [-0.0055, 0.0774]
Waveform3	S	0.3110 $\pm$ 0.0061	0.4716 $\pm$ 0.0111	0.5923 $\pm$ 0.0092	+0.0739 [0.0327, 0.1161]
No dilation	P	0.7127 $\pm$ 0.0017	0.8325 $\pm$ 0.0068	0.9137 $\pm$ 0.0023	-0.0029 [-0.0149, 0.0095]
No dilation	S	0.3282 $\pm$ 0.0080	0.5133 $\pm$ 0.0062	0.6524 $\pm$ 0.0062	+0.0322 [0.0059, 0.0600]
Logistic (n=1)	P	0.7110	0.8117	0.8669	+0.0179 [-0.0055, 0.0445]
Logistic (n=1)	S	0.3791	0.5090	0.6354	+0.0365 [-0.0021, 0.0786]

No two-SD significance threshold is used. Full-minus-probability-only, waveform-only and no-dilation intervals are above zero for S but include zero for P. Full-minus-logistic intervals include zero for both phases. These exploratory findings do not establish equivalence or essential nonlinearity.

Supplementary Table S7. Seed-42 counts and precision/recall at $\pm$ 100 ms, preserving the record-level details underlying the three-seed summaries. Each mode has N=330 records; accepted includes detected and uncertain.

Mode	Phase	TP	FP	FN	Precision	Recall	F1
Full3C	P	276	51	54	0.8440	0.8364	0.8402
Full3C	S	181	141	149	0.5621	0.5485	0.5552
Z-only	P	264	64	66	0.8049	0.8000	0.8024
Z-only	S	90	225	240	0.2857	0.2727	0.2791

Supplementary Table S8. Dataset roles in the two-level workflow. Outer validation is intentionally reused across final model-selection stages. Current test labels are used for scoring, not v2 weight fitting or checkpoint selection; historical roles before correction are documented in Table S23.

Stage	Weight fitting / feature source	Selection / evaluation
OOF bases (each fold)	85% of non-held training sources; augmented fitting records	15% non-held sources for early stopping; held fold excluded from both
OOF meta features	Held-fold originals and augmented copies; matching fold bases	No held source in base fitting or inner validation
Final CNN/TCN	Outer training only; original + augmented	Outer validation for base checkpoint
Neural meta models	OOF features from outer training	Outer validation features from final bases for checkpoint
Logistic comparator	Subsampled OOF training; training-only scaling	Convergence criterion; no validation checkpoint
Weighted ensemble	Phase-specific validation F1 fixes coefficients	Outer validation reused; no test fitting
Decoder	Fixed archived thresholds and S window	No new threshold optimization in diagnostic analyses
Final evaluation	330 test records / 221 sources	Fixed predictions; source-cluster bootstrap and diagnostic scoring

Supplementary Table S9. Accounting of the 40 completed training invocations. One logistic invocation fits two binary heads. Classical pickers and ensemble weighting are not neural training fits.

Training group	Full3C	Z-only	Total
CNN + TCN, five OOF folds	10	10	20
CNN + TCN, final bases	2	2	4
Full15, meta seeds 42–44	3	3	6
Probability12, Waveform3, no dilation; three seeds each	9	0	9
Logistic comparator	1	0	1
Total	25	15	40

Supplementary Table S10. Detected-only results at ± 100 ms on all 330 test records. Mean $\pm$ sample SD over meta seeds 42–44; fixed bases and split. S gating remains based on primary accepted P. Coverage is the mean percentage of finite detected picks.

Mode	Phase	F1	MAE (ms)	Coverage (%)
Full3C	P	0.8337 $\pm$ 0.0065	127.12 $\pm$ 2.04	97.98
Full3C	S	0.5487 $\pm$ 0.0074	480.49 $\pm$ 41.75	94.75
Z-only	P	0.8045 $\pm$ 0.0032	144.92 $\pm$ 20.44	97.88
Z-only	S	0.2606 $\pm$ 0.0130	871.31 $\pm$ 161.18	64.55

Supplementary Table S11. Conditional S diagnostics at ± 100 ms. Each denominator is the stated P-defined subset, not the whole test set. P-rejected and constrained/unconstrained difference counts are also retained in decoder_diagnostics.csv. Unconstrained decoding changes both search domain and quality scoring.

Mode / seed	P subset	N	S F1	S MAE ms
Full3C/42	P accepted	327	0.5578	492.20
Full3C/42	P within 100 ms	276	0.6058	445.92
Full3C/43	P accepted	326	0.5387	617.62
Full3C/43	P within 100 ms	271	0.5885	573.65
Full3C/44	P accepted	329	0.5469	553.51
Full3C/44	P within 100 ms	271	0.5874	512.66
Z-only/42	P accepted	328	0.2799	1170.00
Z-only/42	P within 100 ms	264	0.3217	1090.87
Z-only/43	P accepted	324	0.2825	1220.65
Z-only/43	P within 100 ms	265	0.3095	1182.09
Z-only/44	P accepted	326	0.2885	1150.56
Z-only/44	P within 100 ms	264	0.3198	1081.96

Supplementary Table S12. Network–receiver coverage from recovered upstream metadata. Keys combine network_code and receiver_code and do not identify station epochs. Source grouping does not exclude station reuse or establish geographic generalization.

Subset	Records	Sources	Station keys	Network codes
train	1544	1034	101	31
val	360	222	59	23
test	330	221	64	23

The most frequent test codes are PB (99 records), AV (36), HV (26) and MN (23). PB contributes 30.0% of test records. Geographic coordinates are available in recovered metadata, but no geographic holdout was designed or evaluated. Fifty-five of 64 test network–receiver keys occur in training (318 records). Nine unseen keys contain 12 records from ten sources and are also absent from validation. Table S26 gives retrospective performance strata; these keys do not identify station epochs.

Supplementary Implementation Details

Classical score construction

These definitions describe the retained classical implementations used in the corrected v2 run. Two deterministic score generators and two learned softmax models supply the base features. Classical P/S/noise scores are bounded but uncalibrated and need not sum to one.

STA/LTA uses $fs=100$ Hz, $staSec=0.10$, $ltaSec=1.00$, $trigOn=3.0$, and Gaussian width 0.15 s. For MATLAB indices $i=101, \dots, 6000$, LTA is $\text{mean}(x(i-100:i-1))$ and STA is $\text{mean}(x(i-10:i))$; the latter therefore contains 11 samples. The first 100 ratio values are zero. The ratio is $STA/\max(LTA, 10^{-10})$, followed by record-wise min-max normalization. A range below 10^{-10} yields an all-zero normalized ratio. P uses $\text{abs}(Z)$. S uses $\text{sqrt}(E^2+N^2)$ only when both E and N contain nonzero values; otherwise it uses $\text{abs}(Z)$.

The first global maximum of each normalized ratio determines the score centre. The threshold is $\text{trigOn}/(\text{trigOn}+1)=0.75$ on that normalized curve. Every nonconstant normalized ratio has maximum 1, so it produces a unit-height Gaussian $\exp(-(t-t_{\text{peak}})^2/(2 \times 0.15^2))$; this is not raw-ratio threshold triggering. The alternative branch smooths the normalized curve using smoothdata with a Gaussian window of 15 samples. Scores are clipped to $[0,1]$. $\text{trigOff}=1.5$ is present in configuration but unused; no hysteresis is implemented.

AIC uses signed Z for P and the same horizontal-resultant/fallback rule for S. Candidate indices are $k=2, \dots, \min(T-2, 1000)$. The criterion is $k \log(\max(\text{var}(x(1:k)), 10^{-15})) + (T-k-1) \log(\max(\text{var}(x(k+1:T)), 10^{-15}))$, where MATLAB var uses sample variance. Noncandidate indices are Inf; the earliest finite minimum determines a Gaussian score with $\text{sigma}=0.10$ s and height 1. The right segment extends to the end of the record: this is an initial-index search, not a sliding 10 s window. Time is $(k-1)/100$, so the latest centre is 9.99 s. Both classical generators define $\text{noise}=\max(0, 1-\max(P, S))$.

The AIC centre bound precedes 128/1544 training P and 1148/1544 training S arrivals, 30/360 validation P and 273/360 validation S arrivals, and 15/330 test P and 234/330 test S arrivals. These counts describe centre-search coverage, not final decoded errors. Under Z-only fallback the classical P/S channels are identical by construction; Table S13 verifies this for all current cached meta tensors.

Conditioning and enhanced representation

Each channel is demeaned, linearly detrended, filtered using `butter(4,[1,45]/50,"bandpass")` and `filtfilt`, and normalized by $(x-\text{mean}(x))/(\text{std}(x)+10^{-10})$. Clipping is disabled. An all-zero channel remains zero. Hilbert-envelope channels are divided by their standard deviation only if it exceeds 10^{-10} . Z short-term energy uses $\text{movmean}(Z^2, 10)$, with MATLAB default endpoint handling, and is similarly divided by its standard deviation. The Z characteristic-function channel uses the same 11/100-sample indexing as above, but leaves the ratio zero when $LTA \leq 10^{-10}$; it is divided by its standard deviation only when that deviation exceeds 10^{-10} . No other feature normalization is applied.

Augmentation and fallback

One augmented copy is added per original training record. Operations are additive Gaussian noise (20 dB), one common scale sampled uniformly from $[0.5, 2]$, then horizontal dropout. Noise power is based on mean waveform power across all components in Full3C and Z alone in Z-only; Z-only noise is zero on E/N. With total probability 0.20, dropout selects E-only, N-only, or both uniformly. Each mode has probability $0.20/3$, and each individual horizontal channel is dropped with probability $0.20 \times 2/3$. Time shift and polarity reversal are disabled. E/N are forced to zero again in Z-only. Enhanced features are rebuilt without repeating conditioning; labels are regenerated from the unchanged time grid and arrival times. Neural-layer dropout is separate from this augmentation: CNN 0.30, TCN 0.20, and I-CNN 0.30.

Soft targets and weighted loss

Targets are Gaussian masks with $\sigma_P=0.06$ s and $\sigma_S=0.08$ s, truncated at four standard deviations. $\text{Noise}=\max(0,1-\max(y_P,y_S))$; targets are not normalized. Their sum is $1+\min(y_P,y_S)$. For neural training the per-position contribution is $-\sum_k w_k y_k \log(p_k)$, with weights [1,1,1] for CNN/TCN and [1.5,2.5,0.5] for neural meta-models. Datastore targets are multiplied by the weights before MATLAB trainnet with crossentropy. Its default reduction and the stated L2 regularization are retained. Writing $a_k=w_k y_k$ and $A=\sum_k a_k$, this contribution equals $-A\sum_k (a_k/A)\log(p_k)$: target mass supplies a local weight. It is not crossentropy against normalized targets with uniform sample weight.

A MATLAB R2024a numerical check with predictions [0.2,0.3,0.5] and preweighted targets [1.5,0,0] returned 2.41415691, matching $-1.5 \log(0.2)=2.41415687$. The overlap counts in Table S14 use the original CSV sec grid and the exact truncation expression; rebuilding the grid can alter a boundary sample through floating-point rounding.

Decoder details and verification

The earliest P maximum is selected on the full record. Quality is $\text{peak}/(\text{mean}(\text{score})+10^{-6})$. A candidate is detected for $\text{peak}\geq 0.30$ and $\text{quality}\geq 3$; otherwise it is uncertain for $\text{peak}\geq 0.15$ and not detected below 0.15. Both detected and uncertain P candidates open the S search. Its domain is the record grid 0.1–30 s after accepted P; a numerical tolerance of 10^{-10} s includes endpoints. If P is not accepted or the domain is empty, no S candidate is returned. S quality uses the mean only inside that domain. Unconstrained decoding removes both the gate and interval. Detected-only scoring filters the resulting statuses; it does not rerun the gate using detected P alone.

MATLAB checks passed for zero-waveform normalization, enhanced-feature finiteness, Z-only and one-horizontal-component fallback, preweighted crossentropy, accepted-uncertain P gating, rejected-P gating, and both S-window endpoints. The checks establish current implementation behavior, not the historical rationale for selecting thresholds. The final code and provenance records alone do not establish which data informed their earliest development; no stronger claim about historical threshold selection is made. Documentary threshold evidence is now provided in Table S20; the July values predate the September extension, while the original data-informed rationale remains unknown. Table S27 reports a predefined sensitivity grid applied to fresh fixed-model predictions, without selecting a replacement threshold.

Supplementary Table S13. Complete scan of available Z-only meta-feature caches. Each tensor has $6,000\times 15$ values. Counts refer to cached train, validation, and test features; they do not assert inspection of every temporary historical 8-channel OOF tensor.

Cache	Tensors	NaN/Inf	Nonzero E/N	Unequal STA P/S	Unequal AIC P/S
Augmented train	3088	0	0	0	0
Validation	360	0	0	0	0
Test	330	0	0	0	0

Supplementary Table S14. Overlap of positive P/S target masks on original stored CSV time grids. Augmented copies preserve the labels; doubled counts therefore follow the unchanged-label augmentation rule.

Population	Records	Overlap records	Overlap samples
train	1544	4	53
val	360	0	0
test	330	1	12
Augmented train (labels preserved)	3088	8	106

Availability of implementation checks. Tables S13–S14 were regenerated from the current v2 caches and CSV labels. They are not receipts for historical release caches. The numerical audit scripts and regenerated figures accompany this revision.

Additional base seed experiments and PhaseNetMatched

The extension retains the frozen source split, OOF assignments, augmentation seeds, conditioning, labels, and decoder. Each additional base seed (43, 44) rebuilds its own OOF scores before meta training at seeds 42–44. Base-42 ablations remain restricted to base seed 42. There are 106 fits overall: 40 main, 60 additional stacking fits, and six baseline fits. Repeated meta fits sharing base models must not be counted as independent pipeline replicates.

Supplementary Table S15. Stacking F1@100 ms by base seed. Values are mean $\pm$ sample SD across its three meta seeds; this within-base SD differs from Main Table 6.

Mode	Base seed	P F1	S F1
Full3C	42	0.8296 $\pm$ 0.0093	0.5455 $\pm$ 0.0099
Full3C	43	0.8319 $\pm$ 0.0047	0.5682 $\pm$ 0.0022
Full3C	44	0.8189 $\pm$ 0.0068	0.5472 $\pm$ 0.0129
Z-only	42	0.8059 $\pm$ 0.0041	0.2818 $\pm$ 0.0042
Z-only	43	0.8138 $\pm$ 0.0024	0.2519 $\pm$ 0.0050
Z-only	44	0.8205 $\pm$ 0.0030	0.2721 $\pm$ 0.0161

Supplementary Table S16. PhaseNetMatched per-seed accepted-pick results at $\pm$ 100 ms. All fits use 330 test records; coverage is a percentage.

Mode	Seed	Phase	F1	MAE ms	Coverage %
Full3C	42	P	0.9818	41.09	100.00
Full3C	42	S	0.5788	179.55	100.00
Full3C	43	P	0.9848	28.33	100.00
Full3C	43	S	0.6198	164.00	98.48
Full3C	44	P	0.9833	33.98	99.70
Full3C	44	S	0.5584	166.81	99.70
Z-only	42	P	0.9879	21.48	100.00
Z-only	42	S	0.2939	602.42	100.00
Z-only	43	P	0.9833	26.50	99.70
Z-only	43	S	0.3196	441.99	99.09
Z-only	44	P	0.9788	28.94	100.00
Z-only	44	S	0.2848	468.48	100.00

Supplementary Table S17. Expanded detected-only F1@100 ms and coverage. Stacking averages nine crossed fits; PhaseNetMatched averages three fits. S gating still uses accepted P before applying the detected-only mask.

Method	Mode	Phase	F1	Coverage %
Stack	Full3C	P	0.8306	97.81
Stack	Full3C	S	0.5565	94.58
Stack	Z-only	P	0.8141	97.78
Stack	Z-only	S	0.2571	70.81
PhaseNetMatched	Full3C	P	0.9833	99.90
PhaseNetMatched	Full3C	S	0.5860	98.59
PhaseNetMatched	Z-only	P	0.9838	99.39
PhaseNetMatched	Z-only	S	0.3002	97.88

Supplementary Table S18. Training accounting for the corrected main experiment and extension.

Experiment	Calculation	Fits
Corrected main experiment	Table S9	40
Additional CNN/TCN	2 base seeds $\times$ 2 modes $\times$ 12 fits	48
Additional full stacker	2 base seeds $\times$ 2 modes $\times$ 3 meta seeds	12
PhaseNetMatched	2 modes $\times$ 3 seeds	6
Total	Main plus extension	106

PhaseNetMatched architecture and protocol

PhaseNetMatched follows the encoder–decoder pattern described by Zhu and Beroza (2019), with the SeisBench PhaseNet source as the implementation reference (Woollam et al., 2022; <https://github.com/seisbench/seisbench/blob/main/seisbench/models/phasetnet.py>). It is an independent MATLAB adaptation, not the official implementation. Five levels use 8–128 filters, kernel 7 and stride 4, BN epsilon 10^{-3} , ReLU, transposed convolutions and skip concatenation. Same padding and centre cropping replace explicit asymmetric padding; initialization, biases and BN defaults also differ. No dropout is used. Input is three conditioned waveforms. Training uses batch 2 and the meta-model loss weights [1.5,2.5,0.5]; remaining optimization settings are stated in Methods. Consequently the comparison matches data and evaluation, not feature or compute budgets.

Independent numerical audit

The independent audit recalculated 1,260 metric rows from 70 prediction sets under accepted, strict and unconstrained scoring. Each table contains the same 330 records from 221 verified sources and agrees with the frozen reference labels. Counts, precision, recall, F1, coverage, MAE, median absolute error, RMSE and one-second outlier rates agreed within absolute tolerance $1e^{-6}$ and relative tolerance $1e^{-10}$. All 25,248 entries across current input manifests matched 2,336 unique files. All 106 checkpoints loaded and passed source-exclusion checks. Expanded intervals use 10,000 common source draws with per-fit F1 recomputation before averaging; they exclude training/split uncertainty and are not multiplicity-adjusted. Fresh waveform-to-inference replay is detailed in Table S21.

Archive scope. Source-verified v2 results supersede historical partition assignments and predictions. Release source-verified-v2.0.0 (Git commit 8e82c781a1641191b6b56b15688d68e269b24918) is archived at

<https://doi.org/10.5281/zenodo.22824997>. All 1,241 package-manifest entries, including 106 checkpoints, match the downloaded archive. Earlier release DOIs identify historical states. The archived release retains the earlier manuscript title. This ESI presentation and availability wording postdate the archive; the archived code, protocol, checkpoints, predictions and numerical results are unchanged. The later ESI revision adds a tested retained-CSV reconstruction utility; the new utility and receipts are not inside the existing DOI archive.

Executable software contribution and provenance

Supplementary Table S19. Reusable workflow contracts and executable demonstration. These are integration and auditing capabilities; the underlying grouped-OOF, stacking and hashing techniques are established.

Component	Contract or failure detected	Implementation and scope
Identity propagation	Source identity survives record duplication and augmentation	pipeline_contracts.py; 42 augmented-parent checks
Grouped OOF	Held-out source excluded from fold fitting	grouped_folds / validate_oof; three synthetic folds
Feature interface	Explicit STA/AIC/CNN/TCN P/S/noise plus E/N/Z order	validate_features; makeMeta.m; assertRepresentation.m
Artifact provenance	Checkpoint and waveform identities match recorded hashes	checkpoint_manifest.csv; replay_models.m
Dependency-aware summary	Meta fits are averaged within their shared base before between-base SD	summarize_crossed_seed; archived numerical audit
Model replay	Saved networks reproduce pick times/statuses under the retained decoder	Fresh v2 replay: all 106 checkpoints; test predictions and OOF tensors (Table S21)
Negative controls	Overlap, changed parent, contaminated OOF, swapped features and changed bytes rejected	demonstrate_contracts.py; five expected rejections

Supplementary Table S20. Documentary threshold provenance. Dates below are Git author dates, not evidence of which observations informed initial parameter selection.

Version / commit	Date	Recorded settings
Initial code f566a26b3be5	21 July 2026	peak 0.30; quality P/S 3; S lag 0.1–30 s
Configuration 2a3a87d95a2f	12 August 2026	Same five numerical settings
Original package 767be754d415	15 September 2026	Same settings in original clean configuration
Expanded package e235f749540a	16 September 2026	Same settings; current uncertain threshold = 0.5×0.30

The public v1.0.0 release metadata reports publication on 21 July 2026 at 04:34 UTC. Full commit identifiers, configuration blob hashes and extracted assignments are supplied in threshold_history.json. This establishes documentary continuity of the settings before the September extension. It does not establish validation-only initial calibration or absence of all historical test-informed decisions, and it does not imply identical behavior of legacy and current decoder implementations. No recalibration was performed for this revision. The subsequent 27-setting sensitivity analysis retains the primary configuration and does not resolve the unknown initial calibration history.

Supplementary Table S21. Current numerical, source-verification and fresh inference replay scope and outcomes.

Check	Scope	Outcome
Prediction/label identity	70 sets $\times$ 330 records / 221 sources	All match the frozen test protocol
Metric recomputation	1260 rows; accepted, strict, unconstrained	Zero discrepancies within stated tolerances
Input manifests	25,248 entries / 2,336 unique files	All SHA-256 values match
Checkpoint loading and source audit	106 current fitted models	All load; fit/validation disjoint and test excluded
Actual OOF memberships	6 mode/base manifests $\times$ 1544 records	Match the frozen source-disjoint folds
Fresh test inference	70 sets $\times$ 330 records; two decoders	Zero pick/status mismatches; times within 1e-7 s
Fresh test meta features	6 mode/base groups $\times$ 330 tensors	Maximum absolute difference 0; tolerance 1e-6
Fresh OOF features	6 groups $\times$ 3088 original/augmented tensors	Maximum absolute difference 0; tolerance 1e-6
Replay integrity	106 checkpoint hashes and 89 code files	Unchanged after replay; no training

Fresh GPU inference replay loaded all 106 v2 checkpoints and rebuilt features from the original waveform CSVs. Across six mode/base combinations, all 18,528 original and augmented OOF tensors and 1,980 test meta tensors agreed with their saved counterparts within absolute tolerance 1e-6 (maximum differences 0 and 0, respectively). All 70 final prediction sets, each with 330 records, reproduced constrained and unconstrained pick availability, statuses and times within 1e-7 s, with zero mismatches. Saved validation-derived ensemble weights were retained. This verifies inference and feature reconstruction for the frozen artifacts, not retraining determinism or independent predictive validation. Replay reads the original CSV time grids and waveforms, reapplies conditioning and each input mode, and reconstructs original and augmented OOF inputs in saved order with augmentation seed 4242. Each OOF model excludes its held source from fitting and inner validation. Checkpoint hashes are checked before loading, and all code/model hashes are checked again after completion. Frozen ensemble weights are not re-estimated. Prediction receipts separately report pick/status mismatch counts and numerical quality-score differences. No training is performed.

Supplementary Table S22. Recovery of physical waveform identities and consequences for evaluation. The current run uses recovered source identities and new source-disjoint partitions.

Check	Scope	Outcome
Waveform identity recovery	2234 exported waveforms	2234 numerical matches; 1477 sources
Upstream metadata cross-check	Recovered trace/source identities	2234 agreements
Historical test overlap with train	335 historical test records	151 records share a physical source
Historical test overlap with train or validation	335 historical test records	168 records share a physical source
Corrected outer source overlap	1034 / 222 / 221 sources	Zero in every pair
Original filter claim	Distance ≤ 15 km and min SNR ≥ 10 dB	Not valid for the full retained population
Recovered label convention	CSV labels versus upstream sample	$\text{floor}(\text{sample})/100$ for both phases
Reference pick status	2234 recovered records	1817 both manual; 417 include automatic/autopicker
Exhaustive fingerprint multiplicity	1,265,657 local waveform datasets	2234 distinct selected hashes; one match each
Ambiguous identity / read failures	All selected matches / full archive	0 conflicting sources / 0 read errors

Content matching recovered all 2,234 waveform identities and 1,477 original STEAD source IDs. Upstream metadata agreed for every recovered trace. The earlier ordinal metadata join produced physical-source overlap: 151/335 historical test records shared a source with training, and 168/335 with training or validation. This motivated a complete rerun, rather than relabeling historical model outputs.

The exhaustive audit hashed all 1,265,657 waveform datasets in the local HDF5 archive, with no phase-label or metadata filtering. All 2,234 selected fingerprints were distinct and each matched exactly one trace, with zero conflicting source assignments or read errors. Hashes use little-endian float32 values in column-major E/N/Z order, without normalization. Current CSV fingerprints independently agree with the frozen mapping. This verifies exact numerical multiplicity in this archive version, not near duplicates or catalogue identity truth. Initial identity recovery and split freezing preceded v2 training; exhaustive uniqueness verification followed training. The historical subset-selection procedure remains unavailable; the tested retained-export reconstruction is documented at the end of this resource. All CSV P/S labels equal $\text{floor}(\text{upstream arrival sample})/100$; 94 P and 124 S labels differ from fractional upstream samples by at most 9.01 ms. These labels were retained consistently for all corrected training and evaluation.

Retrospective sensitivity and provenance

Supplementary Table S23. Historical-to-current role transitions. Record counts are exclusive; source counts indicate any membership and are nonexclusive.

Unit	Historical role	Current train	Current validation	Current test
Record	Train	1072	251	233
Record	Validation	249	48	46
Record	Test	223	61	51
Source	Train	804	177	177
Source	Validation	225	46	44
Source	Test	209	53	48

A physical source could occupy several historical roles, so source rows must not be summed. Of 221 current test sources, 196 had at least one historical training/validation record, covering 304 current test records. The remaining 25 sources (26 records) had historical test exposure. This role audit does not reconstruct every research decision or prove that every exposed record influenced one. It establishes no development-naïve test subset. Exclusive source-role patterns and record-level transitions are supplied as CSVs.

Supplementary Table S24. Recovered upstream reference-pick status by current split.

Split	All records	Both phases manual	At least one nonmanual
Train	1544	1224	320
Validation	360	306	54
Test	330	287	43

Nonmanual denotes automatic or autopicker status for at least one arrival. Manual status is a metadata category, not an independent reannotation. The manual-only test sensitivity retains 287 records from 188 sources; the complement contains 43 records from 36 sources. Some sources contribute to both strata. All original model fits are retained.

Supplementary Table S25. S component-access sensitivity at ± 100 ms under accepted scoring. Values average nine stackers or three baseline fits; differences are calculated before rounding. Conditional paired 95% intervals use 10,000 source-cluster draws.

Evaluation subset	Method	N / sources	Full3C F1	Z-only F1	Δ F1 [95% interval]
All test	Stack	330 / 221	.5536	.2686	.2850 [.2293, .3418]
Both manual	Stack	287 / 188	.5476	.2770	.2705 [.2113, .3299]
Both manual	PhaseNetM atched	287 / 188	.5655	.3105	.2550 [.2013, .3085]
No historical train/val	Stack	26 / 25	.4444	.1974	.2470 [.0582, .4462]

Evaluation subset	Method	N / sources	Full3C F1	Z-only F1	ΔF1 [95% interval]
No historical train/val	PhaseNetMatched	26 / 25	.4643	.2451	.2192 [.0503, .3905]
Manual and no historical train/val	Stack	20 / 19	.3647	.1599	.2047 [−.0092, .4390]
Manual and no historical train/val	PhaseNetMatched	20 / 19	.3333	.2342	.0991 [−.0794, .2802]

Manual-subset stacking P F1 is 0.8093/0.7982 for Full3C/Z-only; PhaseNetMatched gives 0.9913/0.9843. Its P advantage remains clear in conditional intervals. Manual S stacker-minus-baseline intervals include zero in both modes. All nine stacking component differences remain positive for manual S (range 0.2403–0.3078). The smallest intersection yields intervals spanning zero for both methods. Neither the subset comparisons nor their overlapping populations isolate a causal annotation effect; the absence of historical train/validation membership does not remove historical test exposure. Full per-fit metrics and intervals for six subsets accompany this table.

Supplementary Table S26. Station-key sensitivity at ± 100 ms with accepted scoring. Network and receiver codes identify keys, not station epochs.

Station-key stratum	Method	N / sources	Full3C P / S	Z-only P / S
Seen in training	Stack	318 / 211	0.8266 / 0.5430	0.8126 / 0.2690
Seen in training	PhaseNetMatched	318 / 211	0.9858 / 0.5742	0.9858 / 0.3024
Unseen in training	Stack	12 / 10	0.8333 / 0.8333	0.8333 / 0.2593
Unseen in training	PhaseNetMatched	12 / 10	0.9167 / 0.8889	0.9167 / 0.2222

The seen stratum contains 55 test keys. The nine keys unseen in training are also absent from validation and include only 12 records from ten sources. This post hoc stratum is not a station-held-out sampling design. For unseen-key S, Full3C-minus-Z-only intervals are [0.3131, 0.7863] for stacking and [0.3636, 0.8974] for PhaseNetMatched. Stacker-minus-baseline S intervals are [−0.1667, 0.0667] in Full3C and [−0.0684, 0.1444] in Z-only. These intervals resample ten sources and do not separately quantify station-level sampling uncertainty. Such a small conditional diagnostic does not support a general station-transfer claim.

Supplementary Table S27. Predefined threshold-grid sensitivity at ± 100 ms. Ranges span 27 settings; counts refer to pointwise exploratory intervals.

Policy / phase	Contrast	Minimum ΔF1	Maximum ΔF1	CI above / below zero
accepted / P	Stack Full3C – Z	0.0127	0.0156	9 / 0 of 27
accepted / P	PhaseNetMatched Full3C – Z	-0.0005	0.0000	0 / 0 of 27
accepted / P	Stack – PhaseNetMatched Full3C	-0.1577	-0.1544	0 / 27 of 27

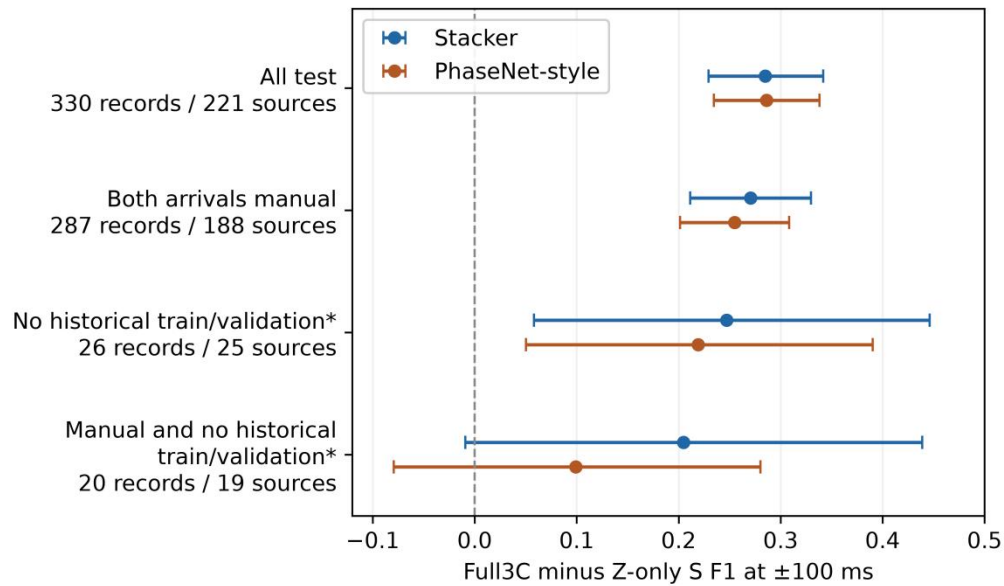
Policy / phase	Contrast	Minimum $\Delta F1$	Maximum $\Delta F1$	CI above / below zero
accepted / P	Stack – PhaseNetMatched Z	-0.1705	-0.1699	0 / 27 of 27
accepted / S	Stack Full3C – Z	0.2839	0.2856	27 / 0 of 27
accepted / S	PhaseNetMatched Full3C – Z	0.2862	0.2873	27 / 0 of 27
accepted / S	Stack – PhaseNetMatched Full3C	-0.0338	-0.0315	0 / 0 of 27
accepted / S	Stack – PhaseNetMatched Z	-0.0316	-0.0305	0 / 0 of 27
strict / P	Stack Full3C – Z	0.0156	0.0168	27 / 0 of 27
strict / P	PhaseNetMatched Full3C – Z	-0.0005	-0.0005	0 / 0 of 27
strict / P	Stack – PhaseNetMatched Full3C	-0.1544	-0.1524	0 / 27 of 27
strict / P	Stack – PhaseNetMatched Z	-0.1705	-0.1696	0 / 27 of 27
strict / S	Stack Full3C – Z	0.2855	0.3031	27 / 0 of 27
strict / S	PhaseNetMatched Full3C – Z	0.2828	0.2875	27 / 0 of 27
strict / S	Stack – PhaseNetMatched Full3C	-0.0325	-0.0265	0 / 0 of 27
strict / S	Stack – PhaseNetMatched Z	-0.0468	-0.0304	0 / 18 of 27

The Cartesian grid varies peak threshold 0.20/0.30/0.40, tied P/S quality threshold 2/3/4 and maximum S–P window 20/30/40 s. Minimum separation remains 0.1 s and the uncertain boundary is half the peak threshold. Eighteen full stacker and six baseline fits are decoded at every setting; the published 0.30/3/30 operating point is unchanged. All configurations, both scoring policies, per-fit metrics and conditional source-bootstrap intervals are supplied as CSVs. Counts are not multiplicity-adjusted tests. Accepted scoring combines detected and uncertain statuses, so quality changes cannot alter its mask or S gate; it reduces to nine peak/window combinations. Strict scoring uses detected candidates but still gates S using accepted P. Changing the S window also changes its quality-score mean. Figure S7 displays only the quality-3 slice; the table and files cover all quality values.

Across the predefined decoder grid, accepted S Full3C-minus-Z-only $\Delta F1$ ranged from 0.2839–0.2856 for stacking and 0.2862–0.2873 for PhaseNetMatched. Pointwise source-cluster intervals were above zero in 27/27 and 27/27 configurations, respectively. Quality-threshold changes leave accepted scoring unchanged by construction; the 27 settings therefore reduce to nine peak/window combinations for accepted scoring, not 27 independent checks. Detected-only S differences ranged from 0.2855–0.3031 and 0.2828–0.2875, respectively. The PhaseNetMatched P advantage retained intervals below zero in both modes at every setting. Accepted S method-comparison intervals included zero throughout, but detected-only Z-only S favoured the baseline in 18/27 pointwise intervals. The stacking P component difference had intervals above zero in 9/27 accepted and 27/27 detected-only configurations, showing dependence on the reporting policy. Supplementary Table S27 reports every contrast summary and Fig. S7 shows the quality-3

slice. These exploratory results neither select a new operating point nor establish historical calibration provenance.

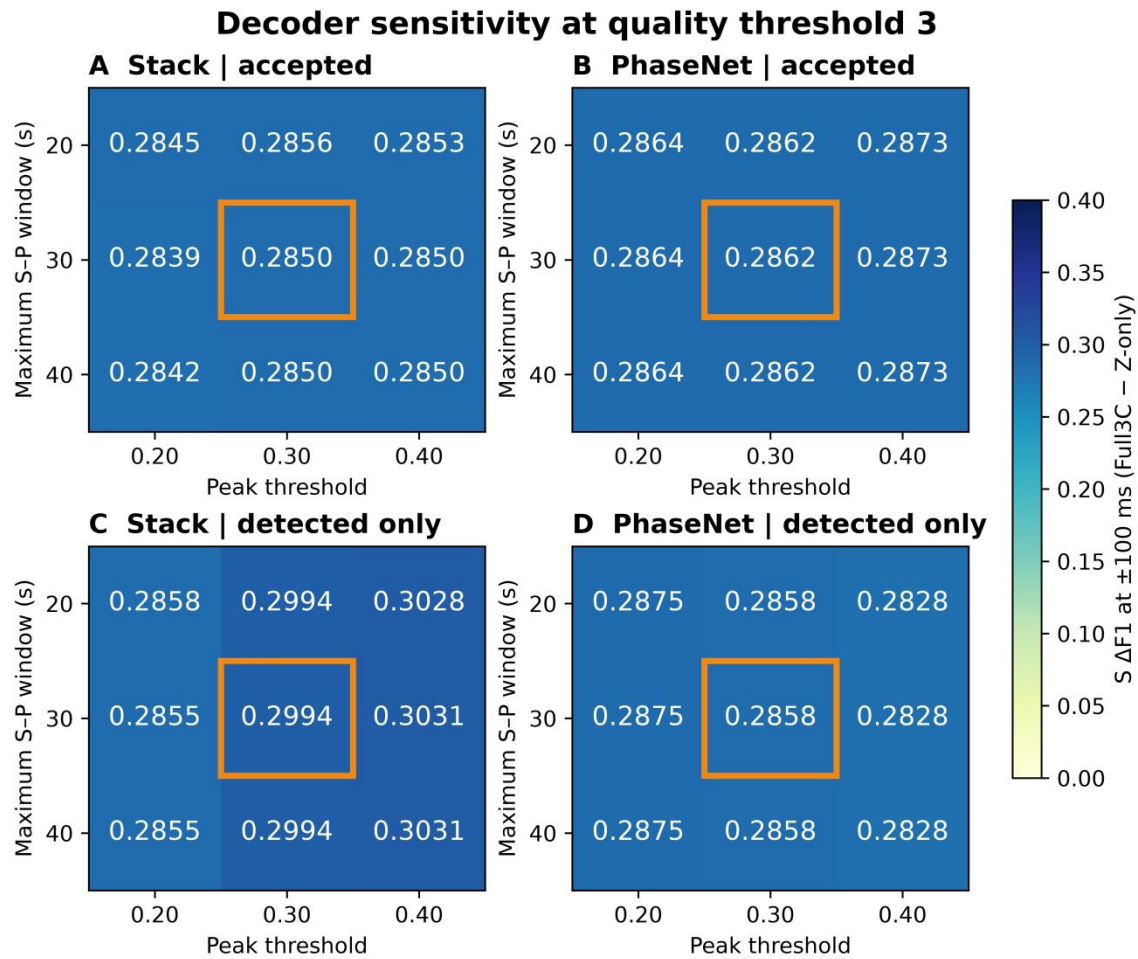
Supplementary Fig. S6. S component-access sensitivity by reference status and historical-role restriction. Source-cluster intervals condition on fixed fits. No subset shown is established to be development-naive.



* These sources occurred in historical test data; no development-naive subset is established.

10,000 source-cluster draws; fixed fits and split. Exploratory, unadjusted intervals.

Supplementary Fig. S7. S Full3C-minus-Z-only F1 sensitivity at ± 100 ms for the quality-threshold-3 slice. Columns vary peak threshold; rows vary maximum S-P window. The original configuration is outlined. Table S27 and accompanying CSVs include all three quality thresholds. No operating point is selected from these results.



Orange outline: retained primary setting.
 Fixed fitted models; no threshold selection.

Additional base-42 methods and diagnostics

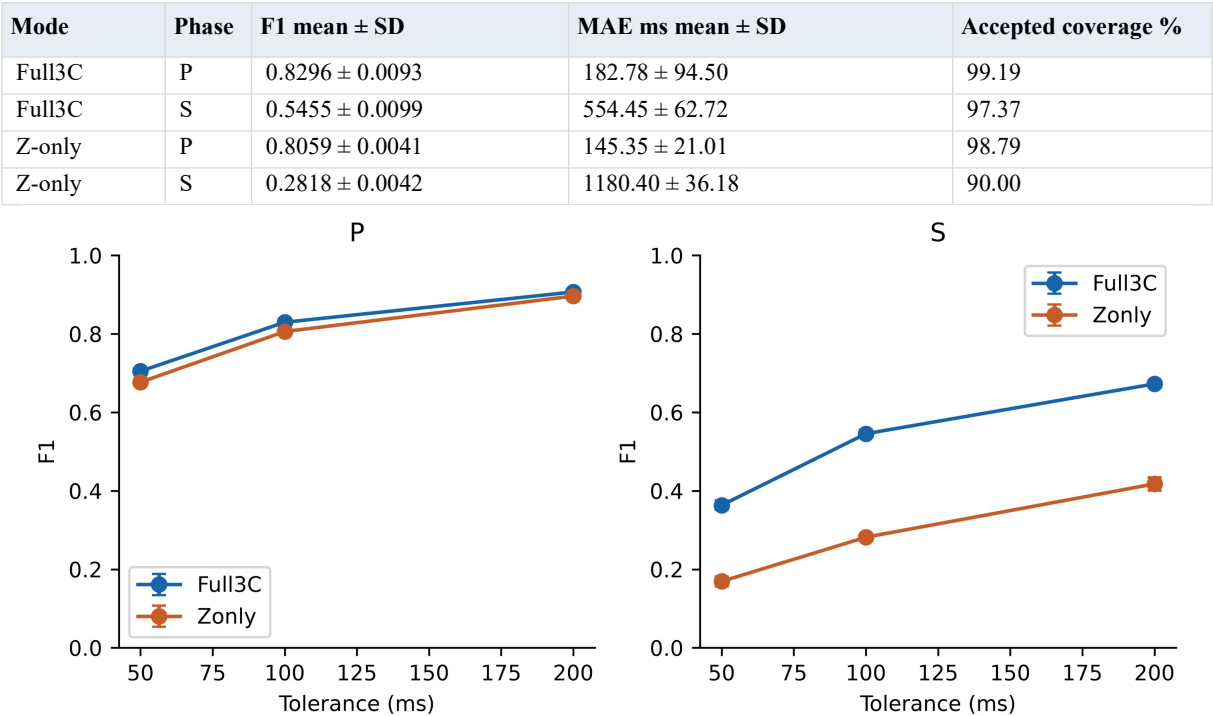
The following tables and figures were moved from the main text without changing numerical values or plotted data. Expanded base/meta results remain the primary presentation in main Tables 6–7; these analyses describe the base-42 experiment only.

Supplementary Table S28. Hyperparameters and configuration settings used for training the CNN, TCN, and I-CNN MetaPicker framework. Base seeds 43–44 belong to the extension; base-42 ablations retain base seed 42. PhaseNetMatched architecture and optimization settings are stated separately in Methods.

Category	Parameter	Final setting
Input	Waveform / base / meta channels	3 / 8 / 15
Input	Sampling; length	100 Hz; 6000 samples
Partition	Outer grouping	source_id; 70/15/15%
OOF	Folds / inner validation	5 / 15% of non-held sources
Conditioning	Band-pass; normalization	1–45 Hz; per component
Labels	Gaussian P / S widths	6 / 8 samples; truncation $\pm 4\sigma$
Base CNN	Filters; kernel; dropout	32,64; 7; 0.30
Base TCN	Filters; kernel; dilation; dropout	32×4; 3; 1,2,4,8; 0.20
Base training	Adam LR; max epochs	1×10^{-3} ; 30
Base training	Batch; validation patience	16; 7 evaluations
I-CNN	Filters; kernel; dilation	64,128,64,64; 5; 1,1,2,4
I-CNN	Dropout; P/S/noise weights	0.30; 1.5/2.5/0.5
Meta training	Adam LR; max epochs	5×10^{-4} ; 50
Meta training	Batch; validation patience	16; 10 evaluations
All neural models	LR schedule; L2	Halve every 20 epochs; 1×10^{-4}
Training	Output network	Best validation loss
Seeds	Base / neural meta / standalone baseline	Base 42–44; meta 42–44 crossed within each base; PhaseNetMatched 42–44. Base-42 ablations: base 42 only.
Augmentation	Training population	Original + one augmented copy
Augmentation	Noise; scale; dropout	20 dB; 0.5–2.0; probability 0.20
Augmentation	Time shift / polarity flip	Disabled / disabled
Decoder	Detected / uncertain	≥ 0.30 and quality ≥ 3 / otherwise ≥ 0.15
Decoder	S after accepted P	0.1–30 s; within record
Evaluation	Primary / strict	Detected+uncertain / detected
Evaluation	Matching tolerances	± 50 , ± 100 , ± 200 ms
Uncertainty	Mean-F1 source bootstrap	10,000 draws; NumPy seed 42

At base seed 42, Full3C mean F1 at ± 100 ms was 0.8296 ± 0.0093 for P and 0.5455 ± 0.0099 for S across three meta seeds. Mean per-run MAE was 182.78 ± 94.50 ms and 554.45 ± 62.72 ms. Accepted coverage averaged 99.19% and 97.37%; it includes uncertain picks and is not recall within 100 ms. Online Resource 1, Table S29 and Online Resource 1, Fig. S8 describe this base-specific result. Section 4.3 reports the expanded comparison.

Supplementary Table S29. Corrected base-42 test performance at ± 100 ms, mean $\pm$ sample SD across three meta seeds (330 records, 221 sources). MAE includes all accepted picks; coverage includes detected and uncertain picks. Expanded results appear in Table 6.

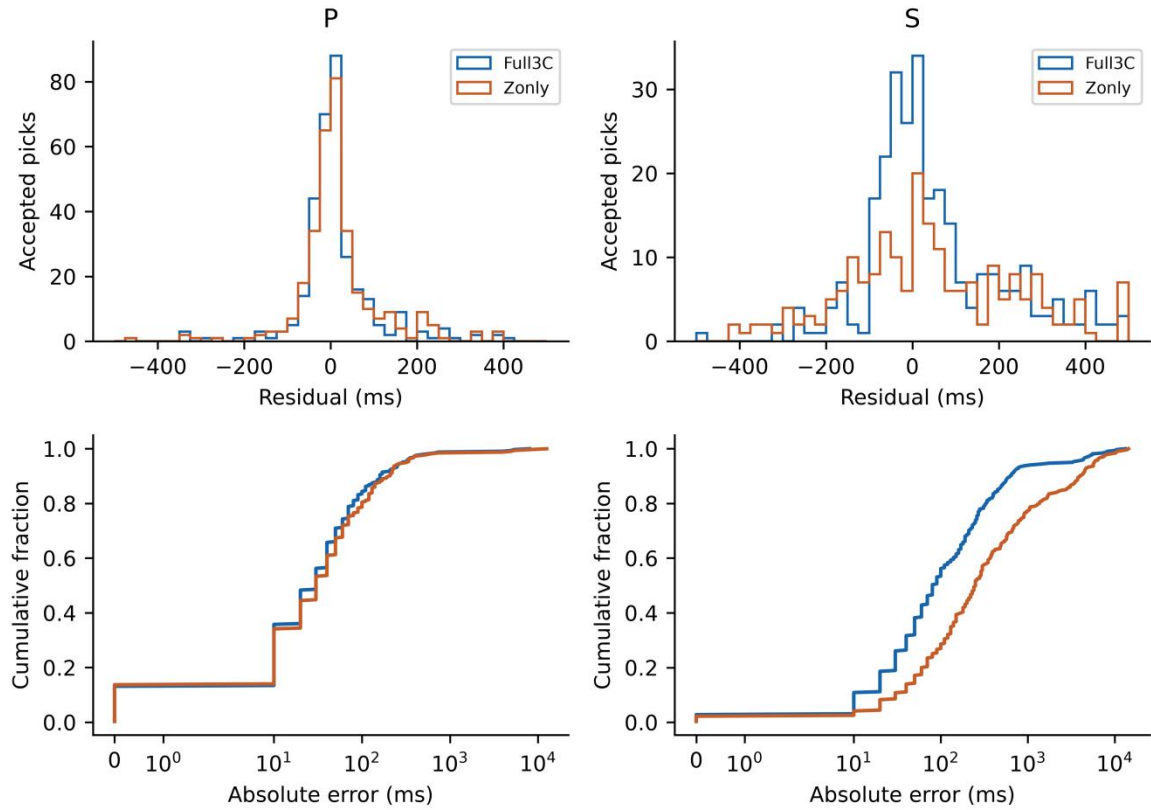


Supplementary Fig. S8. Corrected base-42 F1 versus matching tolerance. Points and bars show mean and sample SD across its three meta seeds. Accepted picks include detected and uncertain outputs.

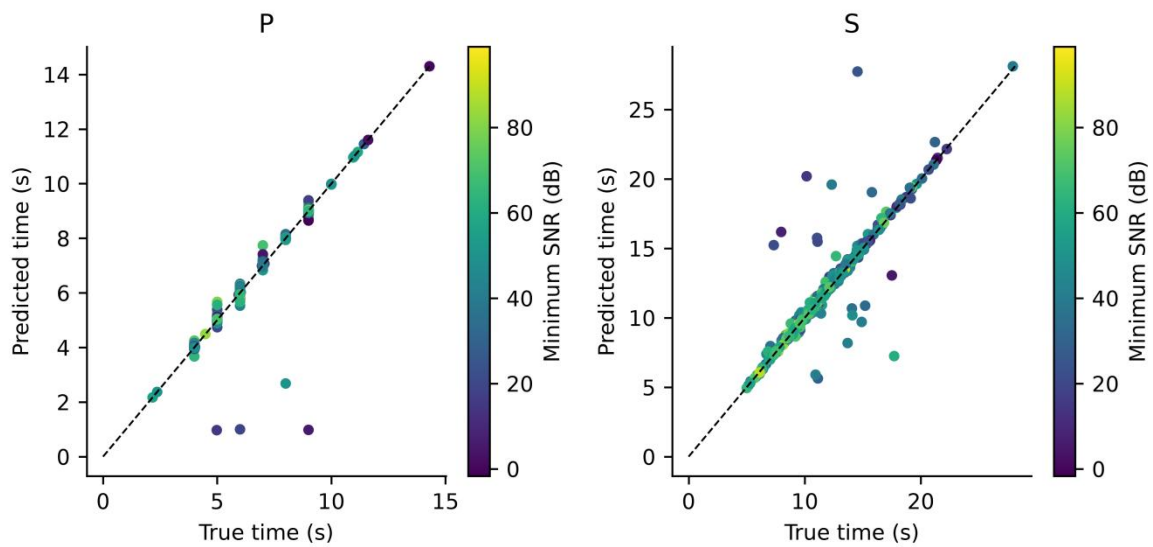
P errors were smaller and F1 higher than S errors and F1. High S accepted coverage coexisted with many picks outside 100 ms: candidate availability is distinct from precise timing. Record-level figures use base seed 42 and meta seed 42; Online Resource 1, Table S29 averages its three meta seeds. These figures are not summaries of the expanded base-seed experiment.

F1 increased with wider tolerance (Online Resource 1, Fig. S8 and Online Resource 1, Fig. S3). Online Resource 1, Fig. S9 presents residual distributions and Online Resource 1, Fig. S10 compares predicted and true times. Bootstrap intervals and detailed errors appear in Online Resource 1, Tables S2 and S3. Tolerance dependence alone does not establish a causal explanation for phase difficulty.

Under detected-only scoring at base seed 42, Full3C P/S F1 was 0.8337/0.5487 versus Z-only 0.8045/0.2606. S detected coverage was 94.75% and 64.55%, respectively (Online Resource 1, Table S10). The S component-access difference therefore persists under the stricter reporting mask. Conditional S metrics and P-rejection counts appear in Online Resource 1, Table S11. Ungated decoding changes both the S search domain and its quality calculation, so it is not a gate-only intervention.



Supplementary Fig. S9. Base-42/meta-42 accepted residuals. Top: histograms within the central ± 500 ms display window. Bottom: full-range cumulative absolute-error distributions on symmetric logarithmic axes. Metrics include errors outside the top display window.

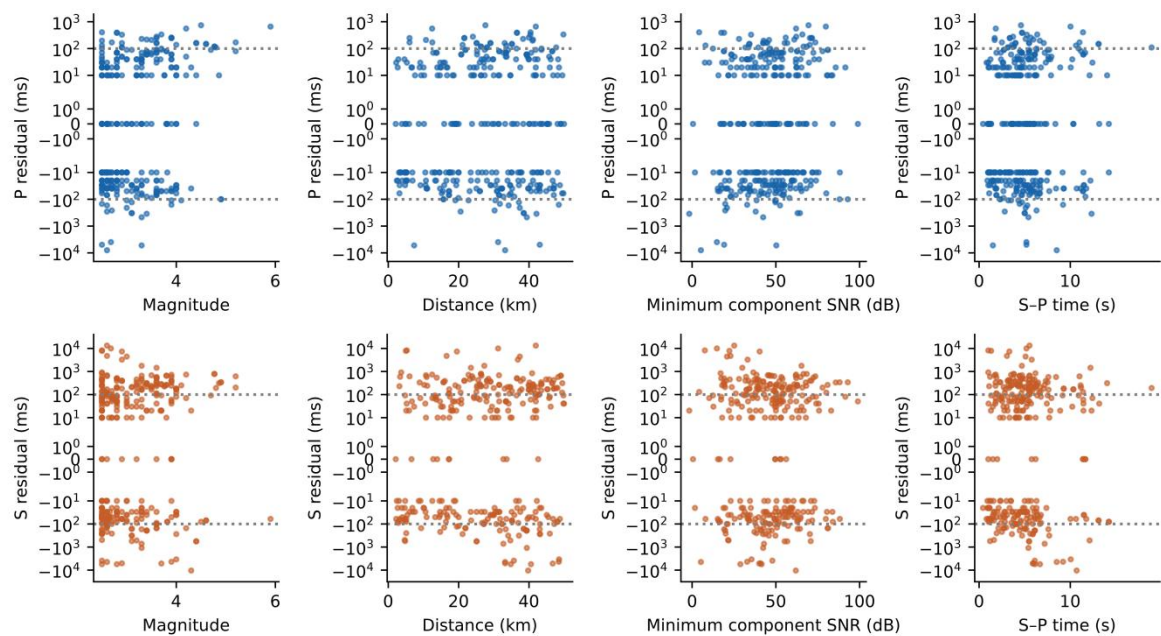


Supplementary Fig. S10. Base-42/meta-42 Full3C predicted versus true P and S arrival times for accepted picks. Colour denotes minimum component SNR; dashed lines show equality.

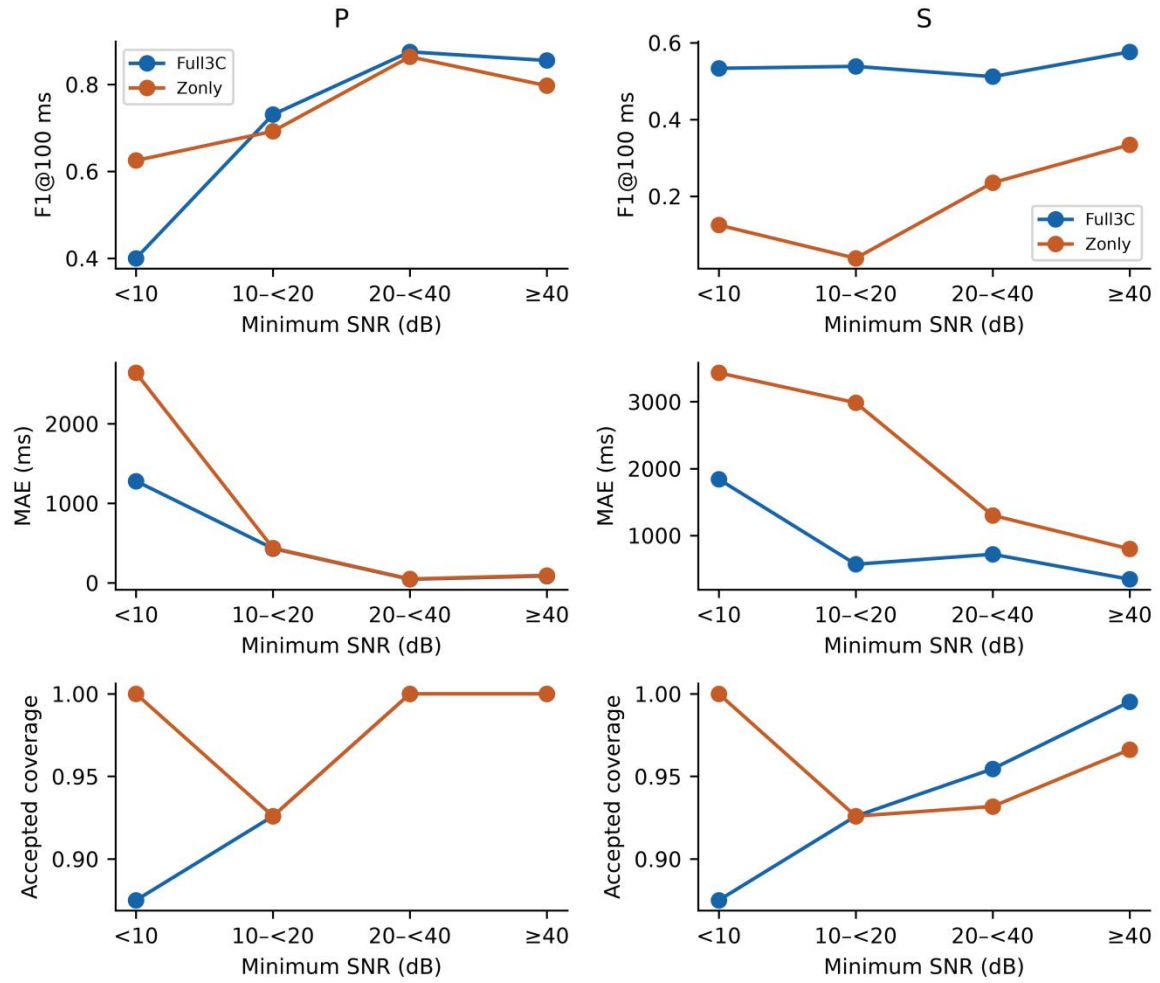
Supplementary Table S30. Corrected base-42 Full3C minus Z-only at ± 100 ms. F1 values average three meta seeds; paired percentile 95% source-cluster intervals use 10,000 draws and condition on those fitted models and the single partition.

Phase	Full3C F1	Z-only F1	$\Delta F1$	95% CI for Δ	Relative change %
P	0.8296	0.8059	+0.0237	[-0.0012, 0.0510]	+2.94
S	0.5455	0.2818	+0.2637	[0.2046, 0.3231]	+93.58

Online Resource 1, Figs. S11 and S12 describe residuals versus metadata attributes and SNR-stratified performance. These analyses did not drive model selection and do not establish causal relationships or controlled noise robustness. Online Resource 1, Fig. S4 ranks the largest P and S residuals.



Supplementary Fig. S11. Base-42/meta-42 Full3C residuals versus distance, magnitude, minimum component SNR, and S–P time. Rows show P and S. Symmetric logarithmic axes retain large errors; dotted lines mark ± 100 ms. Plots are descriptive.



Supplementary Fig. S12. Corrected base-42/meta-42 performance by recovered minimum component SNR: <10, 10-<20, 20-<40 and ≥40 dB. Rows show F1@100 ms, accepted-pick MAE and accepted coverage. Bins are descriptive and were not used for model selection.

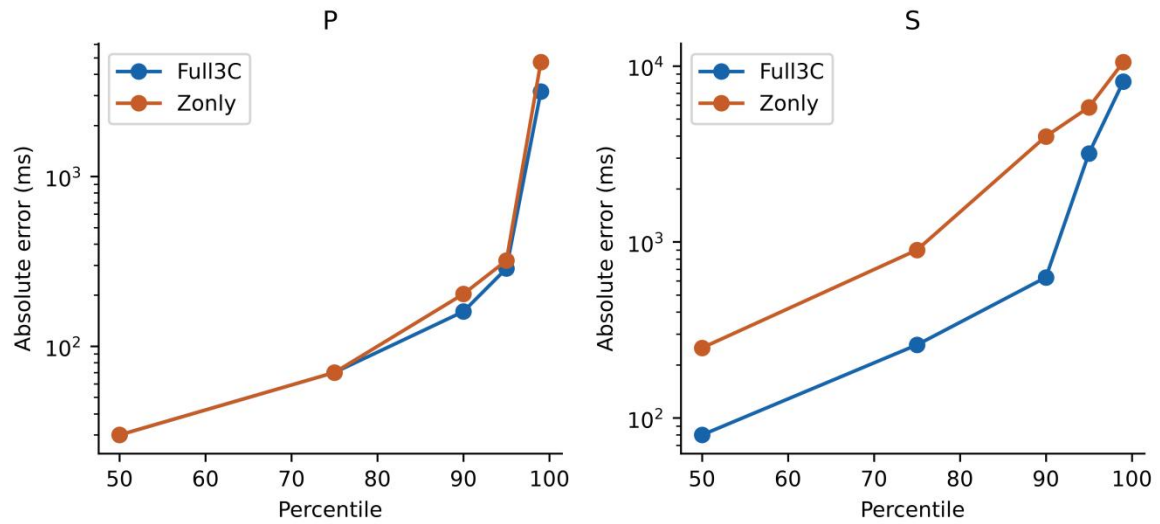
Secondary base-42 stacking and fusion

Within the corrected base-42 comparator set, TCN P/S F1@100 ms was 0.8226/0.4935 versus stacker means 0.8296/0.5455 (Online Resource 1, Table S5). The S difference of 0.0520 had an exploratory paired 95% interval of 0.0165–0.0902. S F1 was 0.1407 for mean averaging and 0.4958 for validation-weighted averaging; the stacker-minus-weighted S interval was 0.0129–0.0885. Weights derive from phase-specific validation F1@100 ms. These conditional comparisons support an S benefit over these particular bases/ensembles, not a general superiority claim.

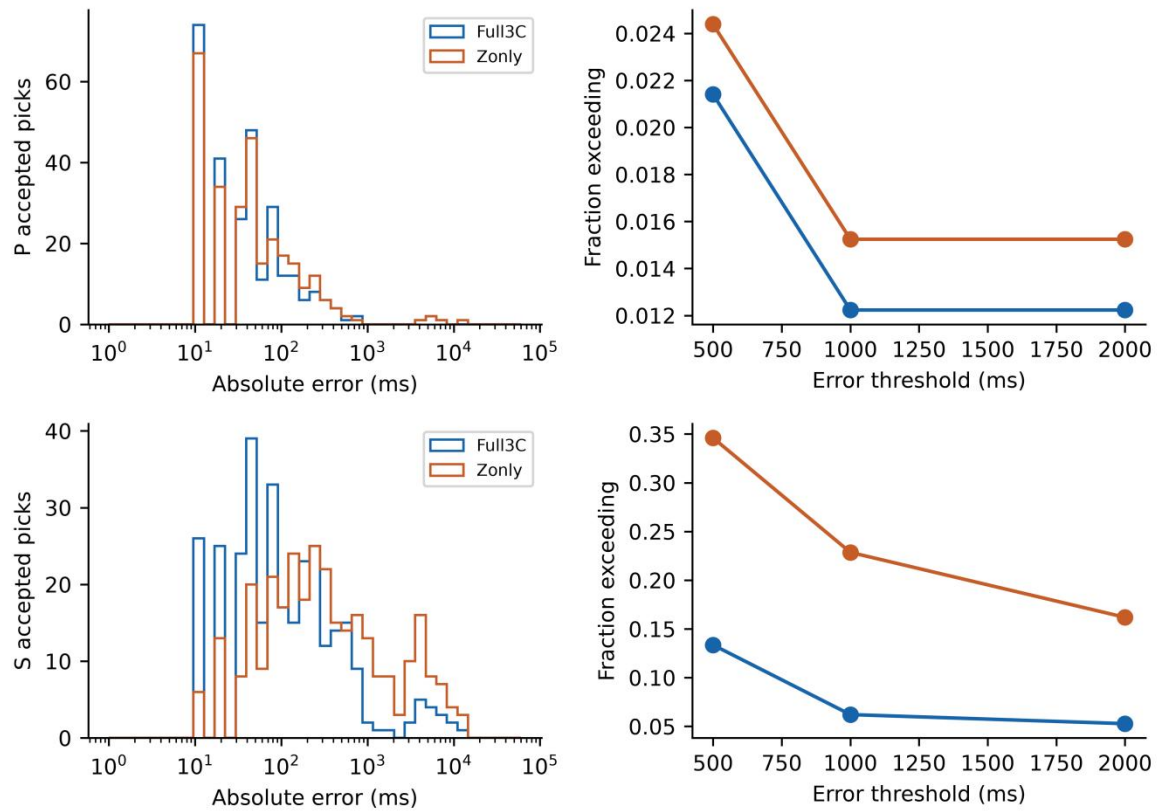
In the corrected base-42 experiment, full 15-channel, probability-only 12-channel, waveform-only 3-channel, and no-dilation meta-learners used seeds 42–44 with shared OOF inputs within Full3C. Online Resource 1, Table S6 reports mean $\pm$ sample SD and paired source-cluster intervals. Seeds vary the meta-learner only; SD is descriptive, not a two-SD significance threshold. Multiple ablation intervals are exploratory.

Probability-only mean P/S F1@100 ms was 0.8327/0.5177, waveform-only 0.7957/0.4716, and no-dilation 0.8325/0.5133. Full-model S differences had exploratory intervals above zero against all three variants, whereas the P intervals included zero. Thus the corrected base-42 ablations support phase-specific gains, including an S dilation benefit on this split, without establishing a universal effect. One logistic stacker achieved 0.8117/0.5090; its full-minus-logistic intervals included zero for both phases. Its independent binary heads use standardized OOF features every tenth sample, ridge penalty 0.01 and targets defined by phase masks >0.3 . This changes temporal context and objective as well as learner type and does not isolate nonlinearity.

Base-42/meta-42 percentile errors are shown in Online Resource 1, Fig. S13; larger accepted errors and selected waveform cases are discussed in Section 5.2.



Supplementary Fig. S13. Base-42/meta-42 accepted absolute-error percentiles (50, 75, 90, 95, 99) for Full3C and Z-only. The vertical axis is logarithmic. Compare error distributions together with acceptance rates.



Supplementary Fig. S14. Base-42/meta-42 accepted errors. Left: full-range histograms on logarithmic axes. Right: fractions exceeding 500, 1000, and 2000 ms. Rows show P and S; denominators are accepted picks.

Retained-input reconstruction verification

The accompanying `regenerate_retained_csv.py` reads the 2,234 trace names in `partition_membership.csv` from the tested STEAD HDF5 archive. It verifies `source_id` and the float32 waveform fingerprint, serializes columns `time`, `sec`, `E`, `N`, `Z`, `p_arrival` and `s_arrival`, and compares the complete output bytes with `waveform_hashes.csv`. The profile uses float32 E/N/Z, float64 `arange(6000)*0.01` seconds, integer-truncated arrival samples multiplied by 0.01, pandas default float serialization, UTF-8 and CRLF line endings. All 2,234 sources, waveforms and CSV hashes matched; errors were zero. Regenerated files are written only on successful hash checks and only to a new output directory. The original selection decisions are not reconstructed. No neural training or new inference was performed for this check.

Environment: Python 3.12.14; NumPy 2.5.3; pandas 3.0.1; h5py 3.16.0. Full hash audit elapsed time was 145.3 s on the author workstation; this is an observed data-regeneration time, not a model-training cost benchmark. Receipts enumerate every retained record and its regenerated SHA-256.

Audit units: 70 prediction sets comprise 36 base/ensemble outputs (six mode/base groups, each with STA/LTA, AIC, CNN, TCN, mean and weighted outputs), 18 full stackers, nine neural ablations, one logistic comparator and six standalone baselines. The 1,260 metric rows are 70 sets × two phases × three tolerances × three scoring policies. The fresh feature replay covered OOF and test tensors; 2,160 outer-validation tensors were not rebuilt. Saved validation-derived weights and selected checkpoints were retained.

Model size and recorded computational time

Supplementary Table S31. Learnable scalar counts and recorded successful-fit wall-clock times, read from all 106 saved checkpoints without training or inference. Fit medians and ranges pool modes, seeds and final/OOF fits; their training-set sizes and stopping times differ. The timing is descriptive, not a controlled efficiency comparison.

Model	Fits	Trainable parameters	Median (min)	Range (min)
CNN	36	16,611	2.99	2.45-6.77
TCN	36	10,467	4.16	3.33-6.18
I-CNN full15ch	18	108,355	12.38	7.00-20.25
PhaseNetMatched	6	269,051	61.76	50.36-74.50
I-CNN probability12	3	107,395	7.77	5.07-12.01
I-CNN waveform3	3	104,515	15.67	14.64-16.15
I-CNN no dilation	3	108,355	9.20	7.85-13.03
Logistic comparator	1	Not neural	1.14	1.14-1.14

Parameter counts sum the elements of the saved network Learnables table, including convolutional biases and batch-normalization scale/offset; running means and variances are excluded. Full3C and Z-only preserve identical channel dimensions and parameter counts. Ablations are Full3C-only. The logistic comparator has two fitted binary heads and is excluded from neural-parameter accounting.

The sum of model.elapsedSeconds is 60,225.14 s (16.73 h) for 106 successful fits. The timer surrounds fitClean and stops before checkpoint serialization. It excludes dataset preparation, OOF feature generation, evaluation, failed/interrupted attempts and user pauses; it is not an elapsed end-to-end experiment duration. No latency, energy or matched-compute advantage is inferred.

No reliable complete replay wall-clock total was retained. The final runner receipt reports 1,923.58 s (32.06 min), but its timer starts after ten groups were complete and covers only Full3C/base44 and Z-only/base42-44 OOF replay, with at most two simultaneous MATLAB processes. This partial duration must not be attributed to all 106 checkpoints or the complete 70-prediction-set replay.

Text and tables use Z-only. Zonly is the unchanged software identifier in paths and legacy plot labels. PhaseNetMatched and PhaseNet-style denote the MATLAB case-study comparator, not the official PhaseNet implementation.